

AMENDMENT NO. 2 SEPTEMBER 2024
TO
IS 17753 : 2021 PAPERBASED MULTILAYER LAMINATED/EXTRUDED
COMPOSITE CARTONS (ASEPTIC AND NON-ASEPTIC) FOR PROCESSED
LIQUID FOOD PRODUCTS AND BEVERAGES — SPECIFICATION

(Page 3, clause 5.2.4) — Substitute the following for the existing and accordingly renumber the subsequent tables:

‘5.2.4 Shelf-Life Test

5.2.4.1 The manufacturer shall declare shelf-life of the paper based multilayer laminated /extruded composite sheet/carton.

5.2.4.2 The paper based multilayer laminated/ extruded composite sheet/carton shall be tested for shelf-life at $(38 \pm 1) ^\circ\text{C}$ and 90 percent ± 2 percent RH (accelerated conditions) or $(27 \pm 1) ^\circ\text{C}$ and 65 percent ± 2 percent RH (standard conditions) as per the schedule given in Table 3.

Table 3 Test Period for the Shelf-life Test

(Clause 5.2.4.2)

Sl No.	Declared Shelf-Life (months)	Accelerated Condition (months)	Standard Condition (months)
(1)	(2)	(3)	(4)
i)	3	1	3
ii)	6	2	6
iii)	9	3	9
iv)	12	4	12
v)	n	n/3	n

NOTES

1 The declared shelf-life (months) is the period from the date ‘date of manufacture’ given at 6.2.2(c).

2 ‘n’ is shelf-life of product in months.

5.2.4.3 Yeast and mould, total bacterial count and total spore count shall meet the requirements as given in Table 4 at the end of schedule of accelerated conditions or standard conditions given in Table 3.

Table 4 Microbiological Parameters for Paper Based Multilayer Laminated/ Extruded Composite Sheet

(Clause 5.2.4.3)

Sl No.	Characteristic	Requirement	Methods of Tests, Ref to
(1)	(2)	(3)	(4)
i)	Total bacterial count, CFU/g, <i>Max</i>	10×10^3	ISO 8784-1
ii)	Total spore count, CFU/g, <i>Max</i>	10×10^3	ISO 8784-1
iii)	Yeast and mould, CFU/100 cm ² , <i>Max</i>	10	ISO 8784-2

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5.2.4.4 The paper based multilayer laminated/extruded composite sheet/carton shall also be visually observed for any delamination and deformation into the laminated structure.

5.2.4.5 The paper based multilayer laminated/extruded composite sheet/carton shall be considered to be conforming when:

- a) The sheet/carton meets the requirements for microbiological parameters at the end of declared shelf-life or at the end of accelerated condition period; and
- b) There is no delamination and no deformation into the sheet/carton at the end of declared shelf-life or at the end of accelerated condition period.'

(Page 3, clause **5.2.5**) — Substitute the following for the existing:

5.2.5 *Water Vapour Transmission Rate (WVTR) and Oxygen Transmission Rate (OTR)*

5.2.5.1 The paper based multilayer laminated/extruded composite sheet passing the shelf-life test shall be tested for water vapour transmission rate (WVTR) according to Annex C or IS 17846 and oxygen transmission rate according to ISO 15105-2.

5.2.5.2 The requirements as agreed between manufacturer and brand owners shall be declared by manufacturer of paper based multilayer laminated/extruded composite sheet based on the **5.2.5.1**. Samples shall be assessed on declared value. The WVTR and OTR when measured in accordance with **5.2.5.1** shall be within ± 5 percent of the declared value.'

(Page 3, clause **6.2.2**) — Substitute the following for the existing:

6.2.2 Each reel comprising of paper based multilayer laminated/extruded composite sheets intended for making composite cartons shall be marked legibly and indelibly with following details:

- a) Indication of the source of manufacture;
- b) Batch number;
- c) Date of manufacture;
- d) Shelf-life; and
- e) Any other statutory requirements.'

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